



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Division: A1

ASSIGNMENT No: 8

TITLE: Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement

Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives

To demonstrate the use of the super keyword for accessing parent class members.

Theory

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

Outcome:

The super keyword was successfully used to invoke parent class constructors, access parent class variables, and call parent class methods that were overridden in the child class, as verified through the sample programs.

```
class Animal {
    String name = "Animal";

    Animal() {
        System.out.println("Animal constructor called");
    }

    Animal(String name) {
        this.name = name;
        System.out.println("Animal parameterized constructor called");
    }

    void sound() {
        System.out.println("Animals make sounds");
    }
}

class Dog extends Animal {
    String name = "Dog";

    Dog() {
        super("Generic Animal");
        System.out.println("Dog constructor called");
    }

    void sound() {
        super.sound();
        System.out.println("Dog barks");
    }

    void showNames() {
        System.out.println("Child class variable (this.name) : " + this.name);
        System.out.println("Parent class variable (super.name): " + super.name);
    }
}

public class JAVA_EXP_8 {
    public static void main(String[] args) {
        Dog d = new Dog();
        System.out.println();

        d.sound();
        System.out.println();

        d.showNames();
    }
}
```

Output:

```
C:\Users\Aashray\OneDrive\Desktop\JAVA_PRACTICALS>java JAVA_EXP_8.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Animal parameterized constructor called
Dog constructor called

Animals make sounds
Dog barks

Child class variable (this.name) : Dog
Parent class variable (super.name): Generic Animal
```

Conclusion:

Therefore, from the experiment we got to learn to use the super key in inheritance.

Exercise

1. Create an Employee and Manager program where Manager accesses Employee details using super keyword.

Source Code:

```
1  class Employee {
2      int empId;
3      String empName;
4      double salary;
5
6      Employee(int empId, String empName, double salary) {
7          this.empId = empId;
8          this.empName = empName;
9          this.salary = salary;
10     }
11
12     void displayEmployee() {
13         System.out.println("Employee ID : " + empId);
14         System.out.println("Employee Name : " + empName);
15         System.out.println("Salary : " + salary);
16     }
17 }
18
19 class Manager extends Employee {
20     String department;
21
22     Manager(int empId, String empName, double salary, String department) {
23         super(empId, empName, salary); // Calling parent constructor
24         this.department = department;
25     }
26
27     void displayManager() {
28         super.displayEmployee(); // Calling parent method
29         System.out.println("Department : " + department);
30     }
31 }
32
33 public class assignment8_EmployeeDemo {
34     Run | Debug | Run main | Debug main
35     public static void main(String[] args) {
36         Manager m = new Manager(empId: 101, empName: "Aashray", salary: 50000, department: "Sales")
37
38         m.displayManager();
39
40     }
```

Output:

```
C:\Users\Aashray\Documents\University\3rd sem\JAVA>java assignment8_EmployeeDemo.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Employee ID   : 101
Employee Name : Aashray
Salary       : 50000.0
Department    : Sales
```

2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

Source Code:

```
1  class Vehicle {
2      String vehicleNo;
3      String ownerName;
4
5      Vehicle(String vehicleNo, String ownerName) {
6          this.vehicleNo = vehicleNo;
7          this.ownerName = ownerName;
8      }
9
10     void displayVehicle() {
11         System.out.println("Vehicle Number : " + vehicleNo);
12         System.out.println("Owner Name      : " + ownerName);
13     }
14 }
15
16 class CarInsurance extends Vehicle {
17     double premium;
18
19     CarInsurance(String vehicleNo, String ownerName, double premium) {
20         super(vehicleNo, ownerName); // Calling parent constructor
21         this.premium = premium;
22     }
23
24     void displayInsurance() {
25         super.displayVehicle(); // Calling parent method
26         System.out.println("Premium Amount : " + premium);
27     }
28 }
29
30 public class assignment8_InsuranceDemo {
31     Run | Debug | Run main | Debug main
32     public static void main(String[] args) {
33
34         CarInsurance car = new CarInsurance(vehicleNo: "MH12AB1234", ownerName: "Amit", premium: 15000);
35
36         car.displayInsurance();
37     }
38 }
```

Output:

```
C:\Users\Aashray\Documents\University\3rd sem\JAVA>java assignment8_InsuranceDemo.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Vehicle Number : MH12AB1234
Owner Name      : Amit
Premium Amount  : 15000.0
```